VNR Vignana Jyothi Institute of Engineering

Computer Science and Engineering | Course Code: CS402

Compiler Design & Program Analysis - Laboratory Manual & Practical Guide

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Syllabus Focus: CS402 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Lexical Analysis & Automata Generation

" Syllabus Scope: Role of lexical analyzer, tokens/patterns/lexemes, regular expressions to DFA Thompsons

Unit 2: Syntax Analysis & Top-Down Parsing

" Syllabus Scope: Context-free grammars, elimination of left recursion, left factoring, FIRST and FOLLOW

CS402: Compiler Design & Program Analysis - Advanced Topics & Assessment

Academic Track: Computer Science and Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Bottom-Up LR Parsing

" Shift-reduce parsing, handles, LR0 items, SLR1 parsing table, Canonical LR1 and LALR1 parsers, Yacc/Bison sp

Unit 4: Syntax-Directed Translation & Intermediate Code

" Syntax-Directed Definitions SDD, synthesized and inherited attributes, Three-Address Code TAC, quadruples, t

Unit 5: Code Optimization & Target Generation

" Basic blocks and flow graphs, local optimizations constant folding, dead code elimination, loop optimization

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Lexical Analyzer Generation using Lex/Flex for C Token Recognition

Lab 2: Recursive Descent Parser Implementation with Syntax Error Reporting

Lab 3: LALR1 Parser Generation using Yacc/Bison for Arithmetic Calculator

Lab 4: Intermediate Code Generation: Synthesizing Three-Address Code from AST

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Alfred V. Aho, Monica S. Lam, Ravi Sethi, Jeffrey D. Ullman - Compilers: Principles, Techniques, and To

Authorized NoteVault Educational Document " Verified for Registered University Students